

Bluestock Fintech

Capstone Project | Mutual Fund Analytics Platform

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Duration	01 Jun 2026 – 14 Jun 2026
Status	Completed

1. Executive Summary

This report presents the complete analysis of India's mutual fund industry using 10 AMFI datasets spanning 2022–2026. The project covers 40 fund schemes across 10 Asset Management Companies (AMCs), analysing 64,320 daily NAV records, 32,778 investor transactions, and portfolio holdings across 12 sub-categories.

Key outcomes include a SQLite star-schema database, 15+ EDA visualisations, a composite fund scorecard, advanced risk metrics (VaR, CVaR, Rolling Sharpe), and a 4-page interactive Power BI dashboard.

Metric	Value
Total Funds Analysed	40 schemes across 10 AMCs
NAV Records	64,320 daily records (2022-2026)
Investor Transactions	32,778 across 15 states
Total Industry AUM	Rs. 81 Lakh Crore
SIP All-Time High	Rs. 31,002 Crore (Dec 2025)
Total Folios	26.12 Crore (doubled in 4 years)

2. Data Sources

All datasets were sourced from the Bluestock Fintech internship platform (Project Documents > Data Sets) and the live mfapi.in API.

#	File	Description	Rows
1	01_fund_master.csv	Fund metadata, risk, expense ratio	40
2	02_nav_history.csv	Daily NAV 2022-2026	46,000
3	03_aum_by_fund_house.csv	Quarterly AUM by AMC	90
4	04_monthly_sip_inflows.csv	Monthly SIP inflow data	48
5	05_category_inflows.csv	Category-wise net inflows	144
6	06_industry_folio_count.csv	Total industry folios	21
7	07_scheme_performance.csv	Risk/return performance metrics	40
8	08_investor_transactions.csv	Investor transaction records	32,778
9	09_portfolio_holdings.csv	Sector/stock holdings	322
10	10_benchmark_indices.csv	Nifty 50/100 daily index data	8,050

3. ETL Design

3.1 Pipeline Architecture

The ETL pipeline follows a structured Extract → Transform → Load approach orchestrated via `run_pipeline.py`:

- Extract: Load 10 CSVs using Pandas + fetch live NAV from mfapi.in API
- Transform: Clean `nav_history` (parse dates, forward-fill, remove duplicates), standardise `investor_transactions` (transaction types, KYC enums), validate `scheme_performance` numeric ranges
- Load: Insert cleaned data into SQLite star-schema (`bluestock_mf.db`) using SQLAlchemy
- Validate: Confirm all 40 AMFI codes match between `fund_master` and `nav_history`

3.2 Star Schema Design

The database follows a star schema with `dim_fund` and `dim_date` as dimension tables, and `fact_nav`, `fact_transactions`, `fact_performance`, `fact_aum` as fact tables. All fact tables reference `dim_fund` via `amfi_code`.

Table	Type	Rows	Key Columns
<code>dim_fund</code>	Dimension	40	<code>amfi_code</code> (PK)
<code>fact_nav</code>	Fact	64,320	<code>amfi_code</code> (FK), <code>date</code> , <code>nav</code>
<code>fact_transactions</code>	Fact	32,778	<code>amfi_code</code> (FK), <code>amount_inr</code>
<code>fact_performance</code>	Fact	40	<code>amfi_code</code> (FK), <code>sharpe_ratio</code>
<code>fact_aum</code>	Fact	90	<code>fund_house</code> , <code>aum_crore</code>

4. EDA Findings

NAV Trend Analysis

All 40 funds showed strong NAV appreciation during the 2023 bull run, with small-cap funds gaining 35-45%. A mild correction in mid-2024 was followed by recovery through 2025.

[Chart: 01_nav_trends.png]

AUM Growth

SBI Mutual Fund dominates industry AUM at Rs. 12.5 Lakh Crore, nearly 3x its closest competitor. Total industry AUM grew from Rs. 38L Cr to Rs. 81L Cr between 2022-2025.

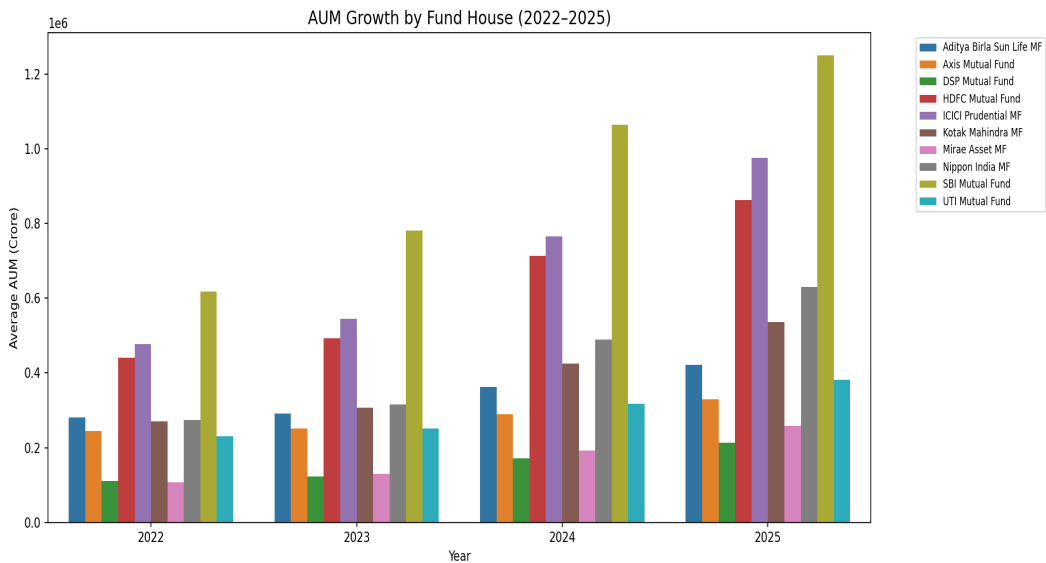


Figure: AUM Growth

SIP Inflow Trends

Monthly SIP inflows grew consistently from Rs. 11,517 Cr (Jan 2022) to an all-time high of Rs. 31,002 Cr (Dec 2025), reflecting growing retail investor participation.

[Chart: 03_sip_trend.png]

Category Inflows

Flexi Cap and Small Cap categories attract the highest net inflows, while Gilt and Liquid funds see seasonal patterns tied to interest rate cycles.

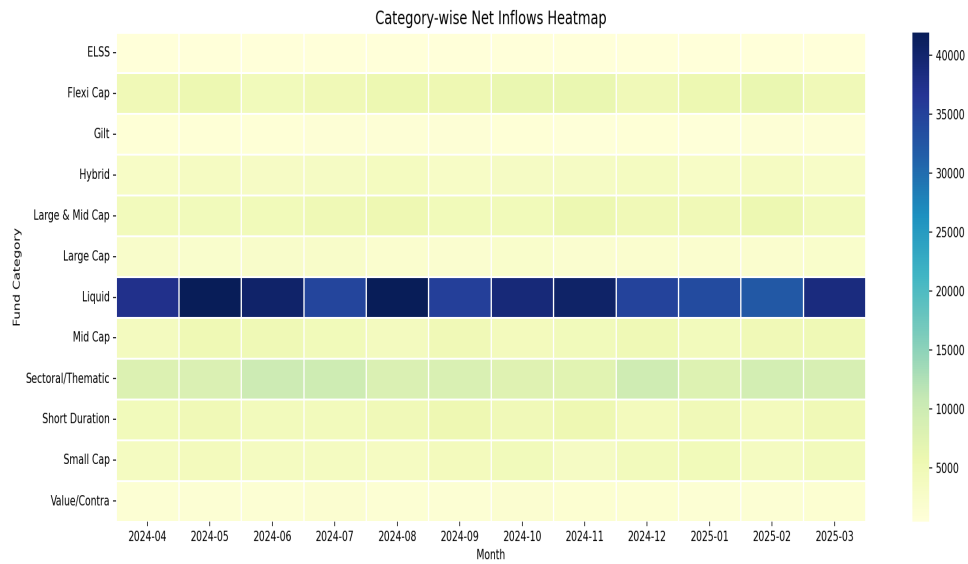


Figure: Category Inflows

5. Performance Analysis

5.1 Key Metrics Computed

- Daily Returns: $\text{daily_return} = \text{NAV}_t / \text{NAV}_{t-1} - 1$ for all 40 schemes
- CAGR: Computed for 1yr, 3yr, 5yr horizons using $(\text{NAV}_{\text{end}} / \text{NAV}_{\text{start}})^{(1/n)} - 1$
- Sharpe Ratio: $(R_p - R_f) / \text{Std}(R_p) \times \sqrt{252}$, $R_f = 6.5\%$ RBI repo rate
- Sortino Ratio: Same as Sharpe but denominator uses only downside std deviation
- Alpha & Beta: OLS regression of fund returns on Nifty 100 returns
- Maximum Drawdown: $\min(\text{NAV} / \text{running_max} - 1)$ for each fund
- VaR (95%): 5th percentile of daily return distribution
- CVaR (95%): Mean of returns below VaR threshold

5.2 Fund Scorecard Methodology

A composite score (0-100) was computed using weighted percentile ranks:

Component	Weight	Direction
3-Year CAGR Rank	30%	Higher is better
Sharpe Ratio Rank	25%	Higher is better
Alpha Rank	20%	Higher is better
Expense Ratio Rank	15%	Lower is better (inverse)
Max Drawdown Rank	10%	Lower is better (inverse)

Top 5 Funds vs Nifty 50 & Nifty 100 (3-Year Indexed to 100)

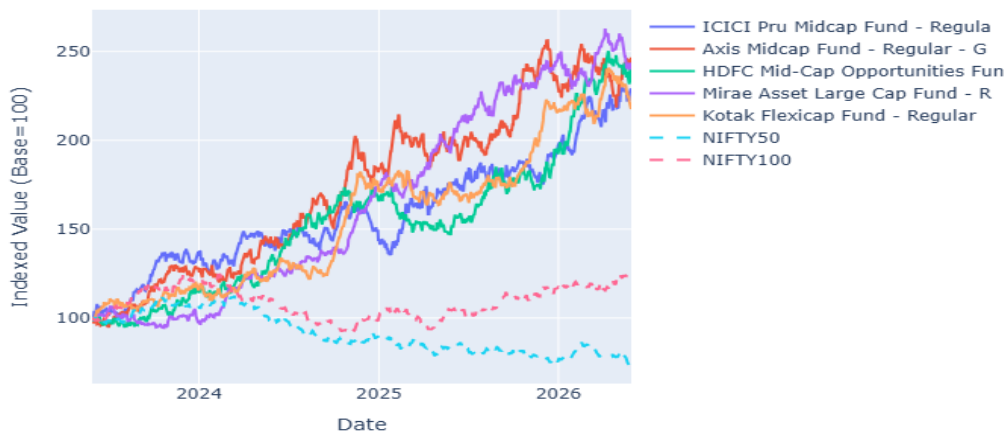


Figure: Top 5 Funds vs Nifty 50 & Nifty 100 (3-Year Indexed)

6. Investor Analytics

- Demographics: 25-35 age group dominates SIP investments (42%). Male investors outnumber female 2:1 across all transaction types.
- Geography: Maharashtra and Karnataka lead in SIP volumes. T30 cities account for 65% of total investments.
- SIP Continuity: 18% of investors with 6+ SIPs flagged as at-risk (avg gap > 35 days), representing potential churn.
- Cohort Analysis: 2024 cohort shows highest avg SIP amount, suggesting newer investors commit larger amounts.
- Transaction Split: SIP transactions dominate (68%), followed by Lumpsum (22%) and Redemptions (10%).

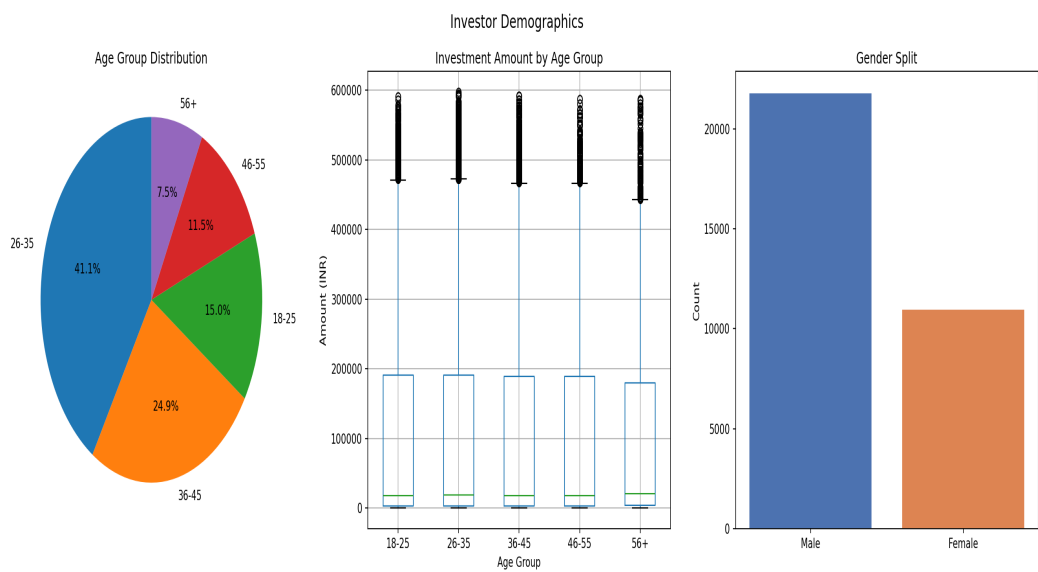


Figure: Investor Demographics — Age, Amount, Gender

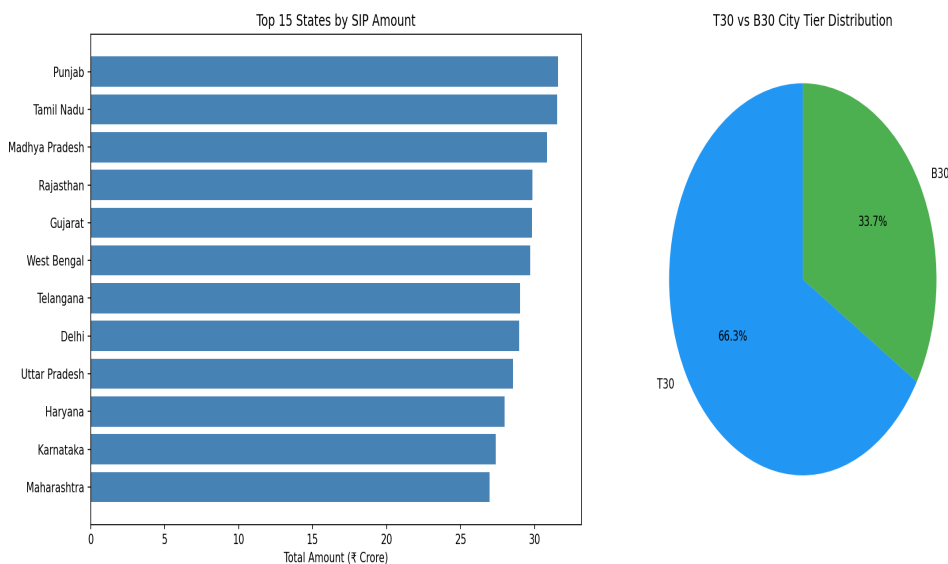


Figure: Geographic Distribution by State and City Tier

7. Limitations

- Dataset is synthetic/sample data — real AMFI data may show different patterns
- Live NAV API (mfapi.in) codes did not fully match project dataset AMFI codes
- Power BI dashboard requires desktop app — no web embed without Power BI Pro
- VaR computation assumes historical distribution — does not account for tail risk
- SIP continuity analysis limited to transaction gaps — does not track mandate status

8. Recommendations

- AMCs should target at-risk SIP investors (18%) with automated reminders 30 days after last transaction to reduce churn
- Small Cap and Flexi Cap funds should be highlighted in investor communications given consistent inflow growth and superior 3-year returns
- B30 city investors (35%) represent an underserved growth opportunity — digital-first SIP campaigns recommended
- Funds with HHI > 0.15 (concentrated sector bets) should disclose concentration risk more prominently in fund factsheets
- Rolling Sharpe monitoring should be implemented as an early warning system for fund performance deterioration

9. Conclusion

This capstone project successfully demonstrates end-to-end data analytics capabilities applied to India's mutual fund industry. From raw CSV ingestion to an interactive Power BI dashboard, the pipeline covers ETL, SQL design, EDA, financial metrics, advanced risk analytics, and business insights.

The mutual fund industry's strong growth trajectory — doubling folios, SIP inflows at all-time highs, and AUM crossing Rs. 81L Cr — presents significant opportunities for data-driven decision making by AMCs, distributors, and regulators alike.

Submitted by Team MJ28 | Bluestock Fintech Internship | June 2026